

Homework 1: likelihood, floating-point arithmetic, and reproducibility

Submission

Submit the following three files in your directory:

1. `lastname_firstname_hw01.ipynb`: one executed Jupyter notebook containing all code and markdown text;
2. `lastname_firstname_hw01.pdf`: a PDF compiled or exported from the executed notebook; and
3. `requirements.txt`: a short list of the packages needed to run the notebook, with version numbers (for example, `numpy==2.1.0`).

Your notebook must run from top to bottom in a fresh session without manual edits.

Question 1: When a likelihood disappears (50 points)

Consider independent observations

$$Y_i \sim \text{Bernoulli}(p), \quad i = 1, \dots, n.$$

Use the following code to generate the data:

```
import numpy as np

rng = np.random.default_rng(123)
n = 20_000
y = rng.binomial(1, 0.03, size=n)
```

1. Write two functions:
 - `likelihood(p, y)`, which evaluates the Bernoulli likelihood as a product; and
 - `log_likelihood(p, y)`, which evaluates the log-likelihood as a sum.

You may assume that $0 < p < 1$. Do not call a prewritten Bernoulli likelihood function.

2. Evaluate both functions on a grid of 400 equally spaced values from $p = 0.001$ to $p = 0.10$. Report how many grid values give a likelihood equal to exactly zero. Explain the numerical cause in two or three sentences.
3. Plot the log-likelihood $\ell(p)$ over the grid and mark both the grid maximizer of the log-likelihood and the analytic maximum-likelihood estimate $\hat{p} = \bar{y}$. Report their values and explain why they may not agree exactly.
4. In two or three sentences, explain why maximizing the log-likelihood is statistically equivalent to maximizing the likelihood but computationally safer.

Question 2: Equivalent formulas, different answers (50 points)

Use the following experiment:

```
rng = np.random.default_rng(123)
x64 = rng.normal(loc=1e8, scale=1.0, size=100_000)
x32 = x64.astype(np.float32)
```

For a vector x , compare the following population-variance formulas:

$$v_{\text{naive}} = \text{mean}(x^2) - \{\text{mean}(x)\}^2, \quad v_{\text{centered}} = \text{mean}\{(x - \bar{x})^2\}.$$

1. Implement both formulas yourself and apply them to `x64` and `x32`. Also compute `np.var(x)` as a reference. Present the six results in a small table showing the method and data type.
2. Treat the generating variance, 1, as a known. For each result, report the absolute error $|\hat{v} - 1|$. Briefly explain:
 - why the naive formula can be inaccurate even though it is algebraically correct; and
 - which discrepancies are numerical error and which remaining discrepancy is statistical error.

Grading guide

Criterion	Points
Correct implementation and results	40
Sound analysis and interpretation	25
Clear figure, table, and concise communication	15
Reproducibility and complete submission	10
Total	100

Minor differences caused by software versions are acceptable. Numerical results without a written explanation will receive limited credit.